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SUPERSEDING
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(Rev. 1) February 2, 1956

FEDERAL SPECIFICATION

POLISH, FURNITURE, WAX

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—Furniture polish covered by this specification is intended primarily for the cleaning and polishing of furniture and other finished wood and metal surfaces.

1.2 **Classification.**—The polish covered by this specification shall be of one grade and one type.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATION

2.1 **Specifications and Standards.**—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

PPP-P-31 — Preservation, Packaging, Packing, and Marking of Material; and Shipping Instructions (Domestic Shipment and Storage).

Federal Test Method Standard:

No. 791—Lubricants, Liquid Fuels and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

JAN-P-108—Boxes, Fiberboard, Corrugated and Solid (Military Type).

MIL-P-10225—Packaging of Glass Containers, One-Gallon Capacity and Smaller, for Other Than Medicinal Products for Domestic Shipment.

Military Standard:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.**—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

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FSC Class 7930

American Society for Testing Materials Specification:

E70–52T—Tentative Method for Determination of pH of Aqueous Solutions with the Glass Electrode.

(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 **Material.**—The liquid furniture polish shall be a stable system of wax and volatile petroleum hydrocarbons suitably emulsified in water. The emulsion shall be a free-flowing fluid that can readily be applied with a cotton cloth and can be easily spread.

3.1.1 The liquid furniture polish shall be free from abrasives and shall be suitable for use on wood and metal furniture finishes. The polish shall have no objectionable odor.

3.2 **Nonvolatile matter.**—Shall be not less than 5 percent nor more than 7 percent by weight when tested as specified in 4.4.1. The nonvolatile matter shall be essentially solid wax-type materials and shall be free of animal, vegetable, or petroleum oils (silicones may be present).

3.3 **Volatile matter.**—Shall consist of water and a volatile petroleum hydrocarbon.

3.4 **Volatile petroleum hydrocarbon content.**—Shall be not less than 30 percent nor more than 40 percent by weight of the sample as received when tested as specified in 4.4.2. The petroleum hydrocarbon shall meet the following requirements when tested as specified in 4.4.3 and 4.4.4:

	Maximum	*Minimum*
Flash point	—	95°F.
Distillation range	420°F.	284°F.

3.5 **Free caustic alkali.**—The polish shall contain no free caustic alkali when tested as specified in 4.4.5.

3.6 **pH of emulsion.**—The pH value shall fall between the limits of 5.0 and 9.5 when tested as specified in 4.4.6.

3.7 **Stability properties.**—There shall be no evidence of settling or breakdown of the emulsion when tested at 43°C. in accordance with 4.4.7.

3.8 **Properties of the film.**—The buffed film shall be clear and practically colorless when tested in accordance with 4.4.8.

3.9 **Abrasive properties.**—There shall be no evidence of abrasive matter in the polish when tested in accordance with 4.4.9.

3.10 **Application.**—The polish shall flow freely and spread easily on the surface to which it is applied when tested in accordance with 4.4.10.

3.11 **Gloss increase and gloss retention.**

3.11.1 The gloss of the panel after polishing shall increase not less than 25 percent (see 4.4.11).

3.11.2 The gloss of the polished panel shall not decrease more than 5 percent when tested in accordance with 4.4.11.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling for lot acceptance.**

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of all furniture polish manufactured as one batch and offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken from each lot by the Government inspector in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level (A.Q.L.) = 2.5 percent defective to verify compliance with all stipulations of this speci-

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fication regarding fill, closure, marking, and other requirements not involving tests.

4.1.3 *Sampling for tests.*—From each inspection lot the Government inspector shall take two containers at random which shall be sealed, marked, and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.2 Inspection.

4.2.1 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined by the Government inspector for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected.

4.3 Lot acceptance tests.—The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4.

4.4 Test procedures.

4.4.1 *Nonvolatile matter.*—Weigh accurately about 10 g. of the sample polish in a tared porcelain crucible or dish. Heat the material over a steam bath until the bulk of the volatile matter is evaporated off, and then heat in an oven at 105° to 110° C. for 24 hours. Calculate the percentage of nonvolatile matter in the polish.

4.4.2 *Volatile petroleum hydrocarbon content.*—To 500 grams of the furniture polish in a 1-liter round bottom flask, add sufficient dilute hydrochloric acid to break the emulsion. Add boiling chips to prevent bumping and connect the flask to a water cooled condenser whose outlet is joined to a 500 ml. flask by a 105° vented connector tube. Ground glass joints are desirable. Heat carefully to distill over the volatile material using a suitable hot plate, electric mantle or oil bath. Separate the petroleum hydrocarbons in the distillate from the water with a separatory funnel and transfer it to a tared beaker. Determine the weight of the petroleum hydrocarbon and compute the percentage. Save the separated hydrocarbon for the determinations in 4.4.3 and 4.4.4.

4.4.3 *Flash point of the volatile petroleum hydrocarbon.*—Determine by method 1101 of Fed. Test Method Std. No. 791.

4.4.4 *Distillation range of volatile petroleum hydrocarbon.*—Determine by method 1001 of Fed. Test Method Std. No. 791.

4.4.5 *Free caustic alkali.*—A 10 ml. sample of the polish shall be digested in 100 ml. of neutral ethyl alcohol (94 percent or higher). Add a few drops of phenolphthalein indicator and if no pink color results at this point, no free caustic is present. Should a pink color appear, dilute about 10 ml. of an original sample of polish with 10 ml. of distilled water, boil and test the vapors for ammonia. If present, continue boiling until the ammonia is driven off. Cool, add 10 ml. of neutral ethyl alcohol (94 percent or higher). Add a few drops of phenolphthalein indicator. No pink discoloration shall result.

4.4.6 *pH of emulsion.*—Determine by ASTM E70–52T using the glass electrode.

4.4.7 *Stability properties.*—A graduated glass cylinder about one inch in diameter shall be filled with a sample of the polish to a height of ten inches. The cylinder shall be glass stoppered and left undisturbed at a temperature of 43°C. for forty-eight hours. The polish shall remain as a uniform emulsion without settling (layer formation) and without breakdown (appearance of solvent on upper surface of the emulsion).

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4.4.8 *Properties of the film.*—Spread a thin film of polish over a clean microscope slide. The slide shall be of clear colorless glass approximately 1.5 mm. thick. Allow the slide to stand in a horizontal position at a temperature of 20° to 30°C. for fifteen minutes. Buff the dried film with a soft dry cloth. Observe the buffed film both by reflected and transmitted daylight. The film by reflected light shall show no signs of whiteness. The film by transmitted light, viewed through the underside of the slide, shall be clear, free from any particles, and practically colorless. The film shall have no odor.

4.4.9 *Abrasive properties.*—Spread a thin film of polish over a microscope slide. Place another microscope slide over the film of polish. Press the two slides together using a slight pressure and at the same time sliding one glass slide over the other. The movement between the glass slides shall be smooth without any indication of the presence of abrasive.

4.4.10 *Application.*—Carry out test under condition of 20° to 26°C. Pour a small amount of polish onto a soft cloth. The polish shall flow freely from the container. Apply the polish with the cloth to a rubbed-varnish wood surface. The polish shall spread easily to a thin film. Allow polish to dry for about two minutes. Buff the film of polish with a soft dry cloth, using no more than moderate pressure. The polished surface shall be nontacky and shall have a high gloss.

4.4.11 *Gloss increase and retention.*—Select a clean rubbed-varnish or rubbed-lacquer panel approximately 1 by 2 feet having a glossmeter reading between 20 and 30 using a Gardner 60° Glossmeter. A minimum of ten readings shall be taken to determine an average, moving and rotating the glossmeter 90° after each reading so that the entire surface shall be represented by the average. This average is the gloss reading of the panel. Apply the polish as directed in 4.4.10, and directly after buffing, determine the gloss reading. The percentage increase in gloss shall be not less than 25 percent. Store the panel for 24 hours at 77°F. and 50 percent ± 2 percent relative humidity and again obtain the average gloss reading to determine gloss retention. The gloss of the polish shall not decrease more than 5 percent.

5. PREPARATION FOR DELIVERY

(For definitions and levels see 6.1)

5.1 Packaging.

5.1.1 *Level A.*—The polish shall be packaged in 1-pint or 1-quart glass bottles (U. S. Standard) as specified in the invitation for bids. Glass containers shall be of good commercial quality and free from defects such as cords, stones, blisters or thin spots which will impair the strength or serviceability. Closures shall be a metal or plastic, continuous thread screw cap. Each cap shall be provided with a liner pad and facing made of materials which will neither affect nor be affected by the product contained, which shall make a leakproof seal effective not only at time of application but during long time storage, and which shall not jar loose in transit.

5.1.2 *Level C.*—The polish shall be packaged in containers in accordance with manufacturer's normal practice.

5.2 Packing.

5.2.1 *Level A.*—Twenty-four one-pint or twelve one-quart bottles shall be packed in accordance with Military Specification MIL-P-10225, except that all interior pads, partitions and liners shall be made from double faced corrugated fiberboard complying with the requirements of Military Specification JAN-P-108, compliance symbol W5c, A or C flute. The shipping container shall comply with the requirements of Military Specification JAN-P-108, Style RSC, compliance symbol V3s or V3c. The size of the individual cells formed by the partitions shall be such as to form a snug fit for the bottle. Shipping

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containers shall be closed and strapped in accordance with the requirements of Military Specification JAN-P-108, except that two straps shall be applied girthwise, over the top, sides and bottom dividing the shipping container into three approximately equal parts.

5.2.2 *Level B.*—Twenty-four one-pint or twelve one-quart bottles shall be packed as specified in Military Specification MIL-P-10225.

5.2.3 *Level C.*—The polish shall be packed in a manner acceptable to common carriers for safe transportation to point of destination specified in shipping instructions, at the lowest transportation rate for such supplies.

5.3 Marking.

5.3.1 *Labeling.*—Each unit container shall bear clear and concise directions for use. The statement "Caution: Combustible mixture" in not smaller than 8 point capital lettters shall appear on the label in a prominent position.

5.3.2 *Civilian agencies.*—Shipping containers shall be marked as specified in Federal Specification PPP-P-31.

5.3.3 *Military agencies.*—In addition to the marking specified in 5.3.1, marking on glass bottles and shipping containers shall be in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Intended use.—The polish is intended for varnished and lacquered, light or dark, furniture finishes and enameled surfaces. It is not intended for exterior surfaces and for floor surfaces.

6.3 Purchasers should exercise any desired options offered herein. (See 5.1, 5.2 and 5.3.)

6.4 Transportation description. — Transportation description applicable to these items is:

Polishing Compounds, not otherwise indexed by name, furniture or vehicle.
Carload weight 36,000 pounds.
Truckload weight 36,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—Q

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